VFD (Variable Frequency Drive) – Workshop Notes

What a VFD Does

A VFD controls the **speed, direction, and torque** of an AC motor by varying the **frequency and voltage** supplied to the motor.

Main Functions

- **Speed Control:** Change motor RPM by varying frequency
- **Direction Control:** Forward / Reverse rotation
- **Acceleration Time:** Smooth motor start (prevents mechanical shock)
- **Deceleration Time:** Controlled motor stopping
- **Multi-Speed Operation:** Preset speeds for different operations

---

Types of Drive Control

- **Internal Control**
  - Start/Stop from VFD keypad
  - Speed set using built-in potentiometer
- **External Control**
  - Start/Stop through push buttons or PLC
  - Speed control via analog signals
- **Both (Internal + External)**
  - Keypad for setup/backup
  - External signals for normal operation

Drive Rating (Speed & Capacity)

- Drive is selected based on **motor HP and current**
- Example:
  - **1 HP Drive → Used for industrial-grade applications**
- Always choose **drive ≥ motor rating** to avoid overload trips

---

Working Principle of VFD (Simple)

1. **AC Supply** enters the VFD
2. Converted to **DC (Rectifier – Unregulated)**
3. **DC Filter** smooths the waveform
4. **Inverter** converts DC back to **variable-frequency AC**

5. Motor speed changes according to output frequency
6. Speed reference given by **potentiometer / 0–10V / 4–20mA**

Common VFD Terminals (Universal)

- **SD:** Common terminal
- **STF:** Forward Run
- **STR:** Reverse Run

Multi-Speed Terminals

- **RL:** Low speed
- **RM:** Medium speed
- **RH:** High speed

👉 **Speed stacking possible**
Example:

- RL → Speed 1
- RM → Speed 2
- RH → Speed 3
- RM + RH → Speed 4
- RL + RH → Speed 5

Analog Input Terminals

Used for **smooth speed control**

- **4–20 mA** (preferred in industry – noise immune)
- **0–10 V**
- **0–50 Hz** (commonly used in Indian electrical systems)

Drive Output & Protection

VFD continuously monitors:

- Motor current
- Voltage
- Load conditions

⚠ **Common Trip Reason:**

- **Overload** (mechanical jam, excess load, undersized drive)

Workshop Tips (Important)

- Always check **motor nameplate data**
- Match **VFD current rating** with motor current
- Set proper **accel/decel time**
- Ensure proper **earthing**
- Use **shielded cables** for analog signals